Healthcare Accessibility in Tripoli, Lebanon

A GIS-Based Spatial Analysis and Interactive Web Mapping Project

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14	2,764	56.1%	75.1%	24.9%
mapped healthcare facilities	buildings analysed	within 500 m	within 1 km	beyond 1 km (687 buildings)

1. Project Overview

Healthcare Accessibility in Tripoli is a Geographic Information System (GIS) project that explores how close buildings in Tripoli, Lebanon are to mapped healthcare facilities (hospitals, clinics and pharmacies).

The project uses QGIS to collect and prepare OpenStreetMap data, build 500 m and 1 km proximity zones around the facilities, classify every building by its distance zone, and present the result as a thematic map. An interactive web map built with Leaflet extends the project so the facilities and distance zones can be explored in a browser.

The project demonstrates a complete GIS workflow: data collection, coordinate reference system handling, spatial analysis, cartographic design and web mapping.

2. Project Objectives

- Map the healthcare facilities within the study area.
- Organise and prepare geographic datasets for spatial analysis.
- Analyse building proximity to healthcare facilities using 500-metre and 1-kilometre buffer zones.
- Visualise geographic proximity through a thematic map.
- Develop an interactive web map for exploring the spatial data.
- Demonstrate practical use of QGIS, spatial analysis and web mapping technologies.

3. Technologies and Tools

Technology	Purpose
QGIS 3.44	Data preparation, spatial analysis, visualisation and the print layout
QuickOSM plugin (Overpass API)	Downloading OpenStreetMap features directly into QGIS
OpenStreetMap	Source of facilities, buildings, roads, the city boundary and the web map basemap
GeoPackage	Storage of prepared datasets and analysis outputs

Technology	Purpose
GeoJSON	Exchange format between QGIS and the web map (EPSG:4326)
Leaflet 1.9.4	Interactive web mapping library
HTML, CSS, JavaScript	Structure, styling and behaviour of the web map
VS Code with Live Server	Development and local testing of the web map
GitHub Pages	Free static hosting for the web map

4. Study Area

The study area is the city of Tripoli in the North Governorate of Lebanon. Its boundary is the OpenStreetMap administrative boundary for the city, retrieved as a single polygon and used to clip the buildings, roads and proximity zones so that all statistics describe the city only.

5. Data Sources

All datasets come from OpenStreetMap and were downloaded in October 2026 with QuickOSM (Overpass API).

Dataset	OSM selection	Geometry	Notes
Healthcare facilities	amenity = hospital, clinic, pharmacy	Points	14 features: 7 hospitals, 5 pharmacies, 2 clinics
Buildings	building = *	Polygons	2,764 buildings after clipping to the boundary
Roads	highway = *	Lines	Clipped to the boundary; used for context
Study boundary	Administrative boundary of Tripoli	Polygon	Selected by its OSM relation identifier

Processing applied: every layer was reprojected to EPSG:32636, buildings and roads were clipped to the boundary, and the three facility types were merged into one point layer with the OSM *amenity* field as the facility type.

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6. Methodology

6.1 Data collection

Features were downloaded with QuickOSM using key/value queries, restricted to the map canvas extent so that results stayed inside Tripoli, Lebanon. The boundary was retrieved by its relation identifier because name searches returned the wrong city or oversized areas. Each result was saved to a GeoPackage immediately, because QuickOSM results are temporary layers.

6.2 Preparation and quality checks

- Checked the CRS, geometry type and attribute table of every layer.
- Reprojected all layers from EPSG:4326 to **EPSG:32636 (WGS 84 / UTM zone 36N)** so that distances are measured in metres.
- Ran geometry validity checks on the buildings and roads.
- Clipped buildings and roads to the Tripoli boundary.
- Merged hospitals, clinics and pharmacies into one facilities layer and checked the feature count (14 = 7 + 5 + 2).

6.3 Buffer analysis

Buffers of **500 m** and **1 km** were created around every facility, dissolved into one shape per distance so that overlaps merge, and clipped to the boundary. The two distances roughly correspond to a 6-minute and a 12-minute walk.

6.4 Building proximity analysis

Select by Location (intersect) was run on the buildings against each buffer. The buildings beyond 1 km are the inverse of the 1 km selection. Selections were exported as separate layers, and the counts were cross-checked: the 500 m count is smaller than the 1 km count, which is smaller than the total, and the 1 km and beyond-1 km counts add up to the total.

6.5 Cartographic visualisation

The final map uses a colour sequence from dark green (within 500 m) through yellow (500 m to 1 km) to red (beyond 1 km), categorised facility symbols by type, dashed buffer outlines, hospital labels, a legend, scale bar, north arrow, title, sources and notes. It was exported as a PDF from the QGIS print layout.

6.6 Interactive web mapping

The prepared layers were exported to GeoJSON in EPSG:4326 with only the attributes the web map needs, and displayed with Leaflet (see section 9).

7. Results

Distance zone	Buildings	Share of 2,764
Within 500 m of a facility	1,550	56.1%
Between 500 m and 1 km	527	19.1%
Within 1 km (cumulative)	2,077	75.1%
Beyond 1 km	687	24.9%

The 500 m to 1 km figure is derived (2,077 minus 1,550).

Key findings

- About **three quarters** of mapped buildings lie within 1 km of at least one mapped facility, and more than half lie within 500 m.
- Roughly **one quarter (687 buildings)** lie beyond 1 km, mostly toward the outer edges of the study area, away from the dense central cluster of hospitals and clinics.
- Facilities are concentrated in the central part of the city, so coverage falls off toward the edges.

These figures describe geographic proximity to *mapped* facilities only. They are not a measure of healthcare quality or of how many people are served.

8. Project Outputs

Output	Description
HealthcareAccess_Tripoli.qgz	Editable QGIS project with styled layers and the print layout
tripoli_healthcare_analysis.gpkg	Clean GeoPackage with the prepared layers and analysis results
HealthCareAccess_Tripoli.pdf	Final thematic map (print layout export)
webmap/	Interactive Leaflet web map: index.html, style.css, script.js and data/*.geojson
README	This document

9. Interactive Web Map

The web map is a static site with no backend or database.

Features

- OpenStreetMap basemap with zoom and pan, fitted to the study area.
- Facility markers coloured by type (hospital, clinic, pharmacy). Popups show the English name, with the Arabic name underneath when available, and the facility type.
- 500 m and 1 km buffer outlines and the study boundary.
- Layer control to switch each layer on or off.
- Legend and an information panel describing the project and its limitations.

The building proximity layers were tested in the web map but left out of the current version because dense polygons covered the facility markers. The building classification is presented in the QGIS map.

Run it locally

Browsers block loading local data files over file://, so serve the folder:

```
cd webmap
python -m http.server 8000
```

Then open http://localhost:8000. Alternatively use the Live Server extension in VS Code.

10. Limitations

This is a basic proximity analysis and should be read within its methodological limits.

- Buffers use **straight-line distance**, not road-network distance or travel time.
- Only **14 facilities** are mapped. OpenStreetMap coverage is uneven, so facilities missing from the data are not reflected and coverage may be understated.
- The buildings layer contains **all** mapped buildings, not only homes, and may be incomplete or out of date. The results are not population figures.
- Pharmacies are counted as facilities alongside hospitals and clinics, although they offer different services.

- Facilities just outside the city boundary can serve residents inside it but are not included.
- Proximity alone does not establish availability, capacity, quality or suitability of healthcare services, and the analysis does not prove that any area is underserved.

11. Reproducibility

Keep the QGIS project, the GeoPackage and the web map together:

```
tripoli-healthcare-gis/
  HealthcareAccess_Tripoli.qgz
  tripoli_healthcare_analysis.gpkg
  HealthCareAccess_Tripoli.pdf
  webmap/
    index.html, style.css, script.js
    data/ (GeoJSON files)
```

1. Open the .qgz in QGIS and confirm that all layers load (project paths are stored relative to the file).
2. Check that the project CRS is EPSG:32636 and inspect the attribute tables.
3. Review the buffer and building-proximity layers.
4. Open the print layout to see the final map.
5. Serve the webmap folder as described in section 9 and open it in a browser.

12. Possible Extensions

- Replace straight-line buffers with road-network service areas or travel-time zones.
- Weight the analysis by population, for example with a population raster.
- Validate the facility list against an official source to reduce the effect of missing OSM data.
- Repeat the analysis for hospitals only, as a sensitivity check.
- Add the building layers to the web map using separate panes so markers stay on top.

13. Project Purpose and Learning Outcomes

This project was developed as a practical application of GIS and web mapping concepts. It demonstrates spatial data management, coordinate reference systems, buffer analysis, vector data visualisation, thematic cartography, geographic data exchange and interactive web mapping.

14. Author

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> **Disclaimer:** This project is intended for educational and portfolio purposes. Its results are exploratory and should not be interpreted as a definitive assessment of healthcare accessibility.